

Cayden J. Osborne

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Education

M.A. Mathematics

August 2026 - May 2027

University at Albany (SUNY), Albany, NY

(Expected)

Thesis: TBA Advisor: Dr. Matthew Zaremsky

B.S. Mathematics (Physics Minor)

August 2023 - Dec. 2026

University at Albany (SUNY), Albany, NY

(Expected)

Major GPA: 3.61/4.00 Overall GPA: 3.56/4.00

Research Interests

Algebraic Number Theory | Geometric Group Theory | Algebra

Research Experience & Projects

Graduate Independent Study

Planned

Department of Mathematics & Statistics, University at Albany, SUNY

Supervisor: Dr. Matthew Zaremsky

- Planned study of Thompson groups, finiteness properties, and the meaning of type F_∞ .
- Will examine CW complexes and group actions as tools for connecting algebraic and topological properties of groups.

UAlbany Showcase Day Presenter

April 2026

Department of Mathematics & Statistics, University at Albany, SUNY

Supervisor: Dr. Anupam Srivastav

- Studied Gaussian Primes and their structure in the ring $\mathbb{Z}[i]$, including all of the characterizations by using norms, and unique factorization. Presented findings at the University-Wide research showcase.

Undergraduate Independent Study

August 2025

Department of Mathematics & Statistics, University at Albany, SUNY

Supervisor: Dr. Changlong Zhong

- Studied fundamental properties of groups, including subgroups, cosets, and quotient groups.
- Explored homomorphisms, isomorphisms, and the isomorphism theorems.

Teaching Experience

- **Substitute Teacher**, Mathematics and Physics

2024 - Present

Instructed classes including AP Calculus, AP Physics, Geometry, Algebra, and Statistics.

- **Private Tutor**, Mathematics and Physics

2022 – Present

Tutored high school mathematics and physics courses, and undergraduate calculus

Coursework

Graduate: Number Theory, Theory of Statistics I, Advanced Linear Algebra, Topology I

Advanced Undergraduate: Abstract Algebra, Probability Theory, Statistics, Computational Mathematics

Relevant Physics: Newtonian Mechanics, Electromagnetism, Modern Physics, Waves and Optics, Computational Physics, and Neutrino Astronomy

Honors & Awards

- Dean's List, University at Albany, SUNY, 2024-2026
- William Roth Scholarship, University at Albany, SUNY, 2025-2027

Skills

Mathematical Software: $\text{\LaTeX}$, Python (NumPy, SciPy)

Programming: Python, Java, C/C++

Languages: English (Native), Spanish (intermediate)